MLOps

Course overview

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Theoretical classes: **it is not mandatory to bring the laptop**.

Practical classes: **you should bring the laptop**, to install and run the several examples that will be given during the classes. Let's try to form small groups to discuss the exercises.

Doubt classes: **Tuesday between 1 and 2 PM** . Students should confirm their attendance by sending an email to the professor. Possibility of doing it on zoom or teams as well on TBD scheduled. But send email whenever you want, I reply :)

Course syllabus theoretical classes

Weeks	Topics	References
21/04 28/04	1-Introduction to MLOps: ❑Motivation; ❑Fundamentals; ❑ Key features. 2-Model Development: ❑Data Unit Tests; ❑Feature Management; ❑Model Training and Experimentation.	Introducing MLOps: How to Scale Machine Learning in the Enterprise 1st Edition Machine Learning Engineering Introducing MLOps: How to Scale Machine Learning in the Enterprise 1st Edition Machine Learning Engineering Beginning MLOps with MLFlow: Deploy Models in AWS SageMaker, Google Cloud, and Microsoft Azur
05/05	3-Model Production and Deployment: ❑ Modular project for CD/CI; ❑ Pipelines orchestration.	https://docs.kedro.org/en/stable/
12/05 19/05	4-Model Deployment requirements: ❑ Model serving; ❑ Containers; ❑ Introduction to Kubernetes and Kubeflow.	Building Machine Learning Pipelines https://www.freecodecamp.org/news/the-kubernetes-handbook/#introduction-to-container-orchestration-and-kubernetes https://www.freecodecamp.org/news/the-docker-handbook/ Kubeflow for Machine Learning
26/05	5-Monitoring and Feedback Loop: ❑ Data and Concept Drift; ❑ Strategies for new Model Deployment.	https://arxiv.org/pdf/2203.11070.pdf https://arxiv.org/pdf/2004.05785.pdf https://docs.seldon.io/projects/alibi-detect/ https://www.nannyml.com/ https://www.evidentlyai.com/
02/06	6-Model Governance: ❑ Explainability and Interpretability; ❑ MLOps and Model Governance.	https://christophm.github.io/interpretable-ml-book/shapley.html

Course syllabus practical classes

Weeks	Topics
21/04 28/04	1-Model Development: ❑Data Unit Tests with **Great Expectations;** ❑Feature Management using **Feature Stores Hopsworks;** ❑Model Training and Experimentation using **MLFlow.**
05/05 12/05	2-Model Production: ❑CI/CD with **Git and Pytest;** ❑Modular code with **Kedro;** ❑Model orchestration with **Kedro.**
19/05	3-Model Deployment and Serving: ❑Containers using **Docker.** ❑**Model Serving** with FastAPI
26/05 02/06	4-Model Monitoring, Explainability and visualization: ❑Detecting Drift**;** ❑**E**xplainability; ❑**Streamlit** for consuming results and present explanations.

Evaluation method

- **Individual examination**: multiple choice questions (**30% of the final grade**).
    1. **1st Exam 02/07/2026 15:00:00**
    2. **2nd Exam 18/07/2026 09:00:00**

- Group project with 3-5 students (**70% of the final grade**):

    1. The aim of the project is to **simulate the real-world process of deploying machine learning models**. More specifically, you need to build a standard and efficient Machine Learning pipeline.

    2. The deliverables of the project should be:

        a) **Project Report**: In at most 6 pages. You should present the definition of the problem, objective, metrics, used technology, risk analysis and conclusions. 10-15 minutes presentation (+ 5-10 minutes of discussion) in the end of the trimester.

        b) **Code and data**: Code associated with the project, a small sample of the data/inputs/outputs if needed, and all steps necessary to replicate your project should be provided along with/in the report. A link to your github is acceptable here (provide it at the fro nt page of the report). The instructions to run it should also be included.

    3. Dealines:

        a. **Until the end of the week 08/05 group constitutions.**
        b. **Week of the 13/05 Midterm PdS with each group**: dataset should be already choosen.
        c. **Final date 30/06.**